Sophia Groom

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Computer Science Student

Computation and Design student at Duke Kunshan University and Duke University with a Computer Science track and experience in software development, technical coursework, team leadership, and project management. Proficient in Python, Java, JavaScript, C, and Scheme, with experience working across China and the United States. Interested in software engineering, AI/ML, and technology-driven problem-solving.

Education & Achievements

Bachelor of Science (BS) | Computation and Design, Computer Science Track| Duke University & Duke Kunshan University Dual Degree| GPA: 3.47 Expected 05/2028

Associate of Science (AS) in University Studies | Utah Valley University | GPA: 3.84 05/2024

Competencies & Technical Skills

Relevant Coursework: Information Science | Programming and Data | Computer Organization | Probability and Statistics Game Development | Discrete Math | Calculus
Programming Languages: Python | Java | JavaScript | Scheme | Lua | C
Standard Office & Collaboration Tools: Microsoft Office | Premiere Pro | Git | Blender | Unity | GitHub | VS Code
Workplace Competencies: Collaboration | Time Management | Cultural Awareness | Team Leadership | Data Analysis Public Speaking | Coaching | Budget Management | Project Planning | Communication
Spoken Languages: English, Native | Chinese, Advanced

Projects

RSA Cryptography Implementation | Python | Racket
- Implemented RSA encryption/decryption, modular arithmetic, and integer factorization including Pollard's rho.

Pico-8 Game | Lua
- Designed and programmed a 2D game in Pico-8, implementing several basic game mechanics.

Computer Architecture Projects | MIPS, C
- Developed programs using MIPS assembly and C, including memory management and data representation.

Roommate Matching Algorithm | Python
- Developed a preference-based matching system comparing student data across eight compatibility criteria.

Leadership Experience

President, Treasurer, & Director | Duke Kunshan University Musical Club Fall 2025 - Present
- Lead a 55-member cast and production team through planning, scheduling, rehearsals, and live performance.
- Manage conflicts and coordinate communication among performers, technical staff, and leadership.
- Plan productions and club activities while overseeing budgeting, purchasing, and organizational operations.

School Ambassador | Duke Kunshan University Fall 2024 - Present
- Represent DKU to prospective students and families by hosting campus tours and answering questions.

Work History

Personal Tutor | Self-Employed 05/2020 - Present
- Provide individualized instruction across multiple subjects, adapting explanations and learning strategies to different students.
- Expanded tutoring services across multiple schools, leveraging word-of-mouth referrals to grow business.